

Data Flow Diagram

Date	25-06-2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how information moves through the Emotion Detection & Learning Support Engine. It shows how user input is processed by the system, how AI models analyse emotions, how Gemini AI generates personalized learning support, and how the results are stored for future analytics. The DFD consists of external entities, processes, data stores, and data flows, providing a clear representation of the system's functionality.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	Represents users interacting with the system, such as Students, Learners, Educators, or Teaching Assistants , who provide study-related input and receive AI-generated learning support.
Rectangle with numbered header	Process	Represents system operations that transform input data into meaningful output, such as Text Preprocessing, Emotion Detection (BiLSTM/BERT), Gemini AI Response Generation, and Result Display .
Rectangle (solid fill, no number)	Data Store	Stores application data for future use, including User Database, Emotion Records, CSV Logs, Model Files, and Analytics Data .
Labelled arrow	Data Flow	Represents the movement of data between users, processes, and data stores, such as Student Input, Processed Text, Predicted Emotion, Confidence Score, AI Response, and Logged Analytics Data .

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

